

Yingrui Hou

Data Scientist | Statistical Modelling, Validation & Monitoring

Clermont-Ferrand, France | +33 (0)785396332 | yingrui.hou@outlook.com

Website: yingrui-hou.github.io/PersonalPage | LinkedIn: linkedin.com/in/yingrui-hou

SUMMARY

Data scientist with a PhD in experimental physics and several years of experience building reusable Python/C++ workflows for statistical modelling, machine-learning validation, calibration, monitoring, and inference on complex measurement data.

Strongest in problems where observations are noisy, ground truth is incomplete, and reliable decisions require dataset alignment, bias correction, uncertainty-aware analysis, and robust validation across operating conditions.

Seeking to transfer this problem-solving experience to industrial data products for manufacturing, process monitoring, and model-driven decision support.

CORE SKILLS

- Data science: Python, NumPy, pandas, scikit-learn, statsmodels, CatBoost / Gradient Boosting
- Modelling: statistical inference, maximum-likelihood estimation, probabilistic modelling, time-series modelling under noise
- ML and data quality: supervised learning, feature engineering, domain adaptation, reweighting, train/test diagnostics
- Validation: model validation, calibration, bias correction, uncertainty quantification, monitoring, anomaly diagnosis
- Engineering: C/C++, Linux, Git, CI/CD, YAML / Shell / CMake automation, large-scale data-processing workflows
- Visualisation and scientific computing: matplotlib, seaborn, ROOT / RooFit, Geant4

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built reusable Python/C++/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where reliable event-level ground truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Developed CatBoost-based selection and validation pipelines with feature studies, train/test diagnostics, threshold scans, and reproducible model application.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.
- Built monitoring workflows that compared expected and observed behaviour, traced anomalies to detector or software causes, and supported calibration updates and corrective actions.
- Worked in Linux-based shared codebases with review discipline, reproducible workflows, and collaboration across large international technical teams.
- Contributed reusable statistical components and uncertainty-stress checks for weighted and unbinned inference workflows on large datasets.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed simulation-backed statistical workflows combining Geant4/C++, signal modelling, calibration, and bias-aware analysis of noisy measurements.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis, component fitting, and calibration on realistic measurement data.
- Quantified a response non-uniformity contribution below 1% in a measurement-to-model validation study and separated physical performance effects from modelling artefacts.

SELECTED PROJECTS

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

Advanced Inference Workflow Validation

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2